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Short Communication

Taxonomic position of new bacteriocin (nukacin ISK-1) producer isolated from long-aged *Nukadoko*

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The new lantibiotic nukacin producers, strain ISK-1 previously labeled as *Pediococcus* sp. ISK-1 and strain Nagano, were characterized biochemically and genetically. Cells were spherical, 0.6 μm in size, and occurred in pairs. The strains were gram-positive, catalase positive and produced more than 75% of DL-lactic acid from glucose and their growth was stimulated by mevalonic acid. The strains tolerated 18% NaCl. G+C content was 36%. 16S rRNA homology revealed that the strains resembled *Staphylococcus warneri.* Both strains were confirmed by quantitative DNA-DNA hybridization to be variant strains of *S. warneri.* These strains have been deposited in the Japan Collection of Microorganism as *S. warneri* JCM 10004 for strain ISK-1 and *S. warneri* JCM 10005 for strain Nagano.

In the northern Kyushu area, many old aged *Nukadoko* (bed of fermented rice bran) have been maintained until now and some are believed to have lasted for a few hundred years (Kimura et al., 1997). *Nukadoko* is usually used for pickle fermentation using washed vegetables without any aseptic treatment. We therefore suspected that *Nukadoko* must possess a native microflora control system that attacks undesirable microorganisms that comes into contact with the bed from the environment. We isolated two organisms, strain ISK-1 and strain Nagano, from an old aged bed owned by Mrs. Sato in Kitakyushu-shi, Japan, a cooking instructor for *Iwashino-Nukamisodaki* which is a local traditional Japanese cuisine, consisting of cooked sardine with old aged *Nukadoko.* These strains were isolated by the following procedure. One gram of *Nukadoko* was suspended in 10 ml of pure water and stood for 24 h at 30°C. One milliliter of the suspension was suspended in 100 ml of thioglycolate (TGC, Difco Laboratories, Detroit, MI, USA) with 1.5% agar and 1.5% $CaCO_3$ to which NaN_3 10 ppm and cycloheximide 10 ppm were supplemented. The agar plates were incubated in an anaerobic jar for 2 days. The organism was purified by repeating single colony isolation on TGC agar plate.

Strain ISK-1 and strain Nagano grew in both anaerobic and microaerophilic conditions. These strains were gram-positive cocci occurring in pairs, catalase positive, and produced more than 75% of DL-lactic acid from glucose with only a little amount of acetic acid. Lactic acid was determined in the form of L- and D- in-

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dividually by HPLC using column SUMICHIRAL OA-5000 (4.6 mmϕ×150 mm, Sumisho, Tokyo, Japan) with a carrier of 1 mM $CuSO_4$ at a flow rate of 1 ml/min. The optical absorbance for the detector was at a wave length of 254 nm. L-Lactate was also confirmed and determined by an enzymatic lactate analyzer model BF-400 (Biott, Tokyo, Japan). Organic acids other than lactic acid and glucose were identified and determined by a Hitachi Model 655A HPLC with a BIO-RAD HPX-87H column and a mobile phase of 0.005 N H_2SO_4. Optimal temperature for growth was 37°C and the strains were tolerant of NaCl at 18%. As displayed in the previous paper, the rate of glucose consumption and lactate formation, namely fermentation time, were tremendously stimulated by the addition of mevalonic acid in the presence of an amino acid solution (Mieki) (Herawati and Ishizaki, 1997).

The G+C mol% was determined by the nuclease P1-HPLC method. DNA of the cell was extracted by Marmur's method and hydrolyzed by P1 nuclease and then alkaline phosphatase according to the method described by Tamaoka and Komagata (1984). Determination was conducted with a Shimadzu Model LC-6A (Kyoto, Japan) HPLC with UV detector whose optical absorbance was at 270 nm. The column was Cosmosil $_5C_{18}$ with a mobile phase consisting of 0.2 M $NH_4H_2PO_4$ and acetonitrile (Ishizaki et al., 1990). Four equimolar mixtures of nucleotides (Yamasa, Chiba, Japan) were used as the standard. The mol% of G+C of the DNA of isolated strains was 36%, which is the same G+C content (36%) as for *S. warneri* ATCC 27836^T.

Strains produced a novel lantibiotic bacteriocin, nukacin ISK-1 (Sashihara et al., 2000), which is sensitive to the growth of *Pediococcus*, *Lactobacillus*, *Lactococcus*, *Micrococcus*, *Clostridium*, and *Bacillus* but not sensitive to strain *Staphylococcus warneri* ATCC 27836^T. These phenotypic characters were not sufficient to assign the strain to a taxonomic position. Thus we have determined 16S rDNA homology and quantitative DNA-DNA hybridization among selected strains genus *Pediococcus*, *Staphylococcus*, and *Micrococcus* for genetic characterization of these strains.

Partial 16S rDNA segments were amplified by PCR, using primers which correspond to nucleotide positions 1101 to 1115 of *Escherichia coli* 16S rRNA (forward primer: 5′-AACGAGCGMRACCC-3′) and to the complement of positions 1392 to 1407 (reverse primer: 5′-GACGGGCGGTGTGTRC-3′) (where R represents A or G and M represents A or C) (Oyaizu, 1992). Partial 16S rDNA segments for positions 8 to 519 were also amplified using our own prepared primers 5′-AGAGTTTGATCCTGGCTCA-3′ and 5′-ATTACCGC-CGCTGCTG-3′. The DNA mainly used for this examination was extracted from the silica scales which formed while the plate was in the tank for 40 days. The 100 µl of amplification reaction mixture contained 0.1 µg of DNA, 10 µl of 10× reaction buffer (500 mM KCl; 100 mM Tris-HCl, pH 9.0; 1.0% Triton X-100), 1.5 mM $MgCl_2$, 0.2 mM of each dATP, dCTP, dGTP, and dTTP, 1 mM of each primer, and 5 U of *Taq* polymerase. The reaction mixture was incubated in a DNA Thermal Cycler (Astec Co., Tokyo, Japan) for 1 min at 96°C, and then subjected to 25 amplification cycles of 30 s at 96°C, 15 s at 58°C and 72°C for 4 min. The amplified fragments were gel-purified on 1% agarose gels, blotted using a DNA Blotting Kit (Takara Shuzo, Kyoto, Japan), and inserted into pUC119 digested with *Sma* I, using a DNA Ligation Kit Ver. 2 (Takara Shuzo). Partial 16S rDNA sequences of randomly selected clones were sequenced using a Sequencing PRO Autosequencer Core Kit (Toyobo, Tokyo, Japan) and a DNA sequencer DSQ-500 (Shimadzu) in accordance with the manufacturer's directions. Three hundred and seven base pairs of the strain corresponding to positions 1101 to 1407 sequences and 512 bp of the strain corresponding to positions 8 to 519 sequences were scanned using the data registered in GenBank by the DNASIS program ver. 3.5. Comparisons of the sequence with GenBank and EMBL data-bases were made using the BLAST network service (Altchul et al., 1990). Sequences of 16S rDNA for the positions from 1101 to 1407 between strain ISK-1 and strain Nagano agreed perfectly and for the positions from 8 to 519 were the same except that one base at the position of 387 for strain Nagano was uncertain. Homology of 16S rDNA analysis for the positions from 1101 to 1407 between isolated strains, ISK-1 and Nagano, and *S. pasteuri*, *S. warneri*, *S. hyicus*, *S. aureus*, *S. hominis*, *S. kloosi*, *S. haemoliticus*, *S. caprae*, *S. carnosus* indicated 99%, while homology for the positions from 8 to 519 has perfectly agreed (100%). However, from this result, it is difficult to determine the species resemblance among *Staphylococcus.*

Quantitative DNA-DNA hybridization was carried out in a microdilution plate as previously described (Ezaki et al., 1989; Kawamura et al., 1998a, b), incubated at 41°C in the solution consisting of 20×SSC (3 M NaCl

Table 1. Determination of DNA-DNA homology of strains ISK-1 and Nagano against the selected strains.

Unlabeled competitive DNA from	Homology value (%)[a]			
	Labeled DNA from			
	ISK-1	Nagano	*S. warneri*	*S. hominis*
Strain ISK-1	100.0	98.0	87.8	18.9
Strain Nagano	95.5	100.0		
Staphylococcus warneri ATCC 27836^T	69.8		100.0	11.0
Staphylococcus hominis ATCC 27844^T	11.5		12.6	100.0
Staphylococcus haemolyticus ATCC 29970^T	10.7		12.7	21.7
Staphylococcus caprae CCM 3573	12.6		15.7	9.5
Staphylococcus capitis ATCC 27840^T	13.4		19.1	13.9
Staphylococcus sciuri ATCC 29062^T	5.2		7.5	5.6
Staphylococcus simulans ATCC 27848^T	6.3		7.3	7.6
Staphylococcus xylosus ATCC 29971^T	8.8		9.5	7.9
Staphylococcus auricularis ATCC 33753^T	8.5		8.8	7.2
Staphylococcus cohnii ATCC 29974^T	6.0		8.2	5.7
Staphylococcus carnosus DSM 20501	9.8		11.7	9.3
Staphylococcus chromogenes NCTC 10530^T	7.6		8.1	9.2
Staphylococcus gallinarum CCM 3572	5.8		10.4	9.0
Staphylococcus hyicus ATCC 11249^T	6.0		9.9	8.4
Staphylococcus lugdunensis ATCC 43809^T	7.8		16.1	10.9
Staphylococcus schleiferi ATCC 43808^T	5.5		13.6	10.3
Staphylococcus kloosii DSM 20676	8.9		16.9	10.3
Staphylococcus saprophyticus ATCC 15305^T	9.6		12.2	14.7
Staphylococcus arlettae DSM 20672	7.0		7.7	9.0
Staphylococcus equorum DSM 20674	5.4		13.4	10.5
Staphylococcus epidermidis ATCC 14990^T	5.6		5.6	4.1
Staphylococcus aureus ATCC 12600^T	7.2		4.7	5.1
Staphylococcus delphini DSM 20771	4.0		7.3	3.4
Staphylococcus intermedius ATCC 29663^T	2.8		11.0	7.0
Staphylococcus caseolyticus ATCC 13548^T	2.6		1.7	1.0
Staphylococcus lentus ATCC 29070^T	1.3		2.1	0.9
Staphylococcus mucilaginosus CCM 2417	1.2		0.7	0.0
Micrococcus varians ATCC 15306^T	0.7		0.0	0.0
Micrococcus luteus JCM 1464^T	2.1		0.0	0.0
Pediococcus damnosus JCM 5886^T	1.5		2.2	2.3
Escherichia coli ATCC 12435	0.0		0.0	0.0

Abbreviations: ATCC, American Type Culture Collection; CCM, Czechoslovak Collection of Microorganism; DSM, Deutsche Sammlung von Mikroorganismen; JCM, Japan Collection of Microorganisms; NCTC, National Collection of Type Cultures.

[a] Homology value (%) is indicated as mean value of two experiments.

and 0.3 M trisodium citrate) 1 ml, 100×Denhardt 0.5 ml, denaturated sperm DNA 0.1 ml, formamide 5 ml, dextran sulfate 0.25 ml and sterilized deionized water 3.4 ml. Strains used were *S. warneri* ATCC 27836^T, *S. hominis* ATCC 27844^T, *S. haemolyticus* ATCC 29970^T, *S. caprae* CCM 3573, *S. capitis* ATCC 27840^T, *S. sciuri* ATCC 29062^T, *S. simulans* ATCC 27848^T, *S. xylosus* ATCC 29971^T, *S. auricularis* ATCC 33753^T, *S. cohnii* ATCC 29974^T, *S. carnosus* DSM 20501, *S. chromogenes* NCTC 10530^T, *S. gallinarum* CCM 3572, *S. hyicus* ATCC 11249^T, *S. lugdunensis* ATCC 43809^T, *S. schleiferi* ATCC 43808^T, *S. kloosii* DSM 20676, *S. saprophyticus* ATCC 15305^T, *S. arlettae* DSM 20672, *S. equorum* DSM 20674, *S. epidermidis* ATCC 14990^T, *S. aureus* ATCC 12600^T, *S. delphini* DSM 20771, *S. intermedius* ATCC 29663^T, *S. case-*

Table 2. Carbohydrate assimilation test for strain ISK-1 and strain Nagano using API CH kit.

Carbon sources	Results at	
	24 h	48 h
Control	0	0
N-Acetyl glucosamine	0	0
Adonitol	0	0
Amidon	0	0
Amygdalin	10	30
D-Arabinose	0	0
L-Arabinose	0	0
D-Arabitol	0	0
L-Arabitol	0	0
Arbutin	0	0
Cellobiose	50	80
Dulcitol	0	0
Erythritol	0	0
Esculin	0	0
D-Fructose	100	100
D-Fucose	0	0
L-Fucose	0	0
Galactose	10	50
β-Gentiobiose	50	50
Gluconate	0	0
D-Glucose	100	100
Glycerol	0	0
Glycogen	0	0
Inositol	0	0
Inulin	20	20
2-Ketogluconate	0	0
5-Ketogluconate	n.t.	n.t.
Lactose	0	0
D-Lyxose	0	0
Maltose	0	0
Mannitol	10	30
D-Mannose	0	0
Melezitose	0	0
Melibiose	50	100
α-Methyl-D-glucoside	0	0
α-Methyl-D-mannoside	0	0
β-Methyl-xyloside	0	0
Rhamnose	0	0
D-Raffinose	0	0
Ribose	20	90
Saccharose	100	100
Salicin	0	0
Sorbitol	0	0
L-Sorbose	0	0
D-Tagatose	0	0
Trehalose	0	0
D-Turanose	0	0
Xylitol	0	0
D-Xylose	0	0
L-Xylose	0	0

Figures in the table indicate the degree of assimilation with color change of the plate as 0 for no change, 50 for green, and 100 for yellow. n.t. means not tested. Other figures indicate the color change evaluated by the above standard.

olyticus ATCC 13548^T, *S. lentus* ATCC 29070^T, *S. mucilaginosus* CCM 2417, *Micrococcus varians* ATCC 15306^T, *M. luteus* JCM 1464^T, *Pediococcus damnosus* JCM 5886^T, and *E. coli* ATCC 12435. Quantitative DNA-DNA hybirdization among the selected strains clearly indicated that strain ISK-1 and strain Nagano were genetically related to staphylococci but not related to any other genus. The homology value between the type strain of *S. warneri* ATCC 27836^T and ISK-1 was in a range from 69.8 to 87.8% under optimal conditions (Table 1). Homology value between strain Nagano and ISK-1 was more than 95%. So we identified both strains as *S. warneri.*

For further confirmation, strain ISK-1 and strain Nagano were biochemically characterized using a Nissui *Staphylococcus* identification kit S-18 profile for N-ID test, and carbohydrate assimilation test by API 50 CH (bioMerieux, Lyon, France) (de Lajudie et al., 1994). Results are shown in Table 2. The profile of carbohydrate assimilation of strain ISK-1 was compared to that of *S. warneri* and there was perfect agreement according to the Bergey's Manual (Sneath et al., 1989). This was also confirmed by N-ID test using a kit for *Staphylococcus* with SP-18 profile. Strain ISK-1 and Nagano agreed with *S. warneri.*

Enterotoxin was detected by TST-RPLA "SEIKEN," a reagent kit for detecting *Staphylococcus* TSST-1 (Toxic Shock Syndrome Toxin) and SET-RPLA "SEIKEN" Staphylococcal enterotoxin A, B, C, and D detection kit by reversed passive latex agglutination (Denka Seiken Co., Ltd., Tokyo, Japan) (Toma et al., 1999). None of the enterotoxins were detected. For the hemolysis test, bacteria were grown on a BHI agar base (Difco Laboratories) supplemented with 10 mM $CaCl_2$ and 5% defibrinated sheep blood (Nissui, Tokyo, Japan) (Beutin, 1989). The strains were precultured in MRS and transplanted onto plates and incubated at 37°C for 24–48 h and observed. Weak hemolysis was observed on the sheep blood plate after 48 h culture.

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